



HR Analytics Project

Analyzing employee data to understand attrition patterns, salary distribution, and workforce behavior for data-driven HR decision-making.

Dataset Overview



Department & Job Role

Organizational structure and position details



Demographics

Age, gender, and education field information



Compensation

Monthly income and salary distribution



Tenure & Experience

Years at company and total working years



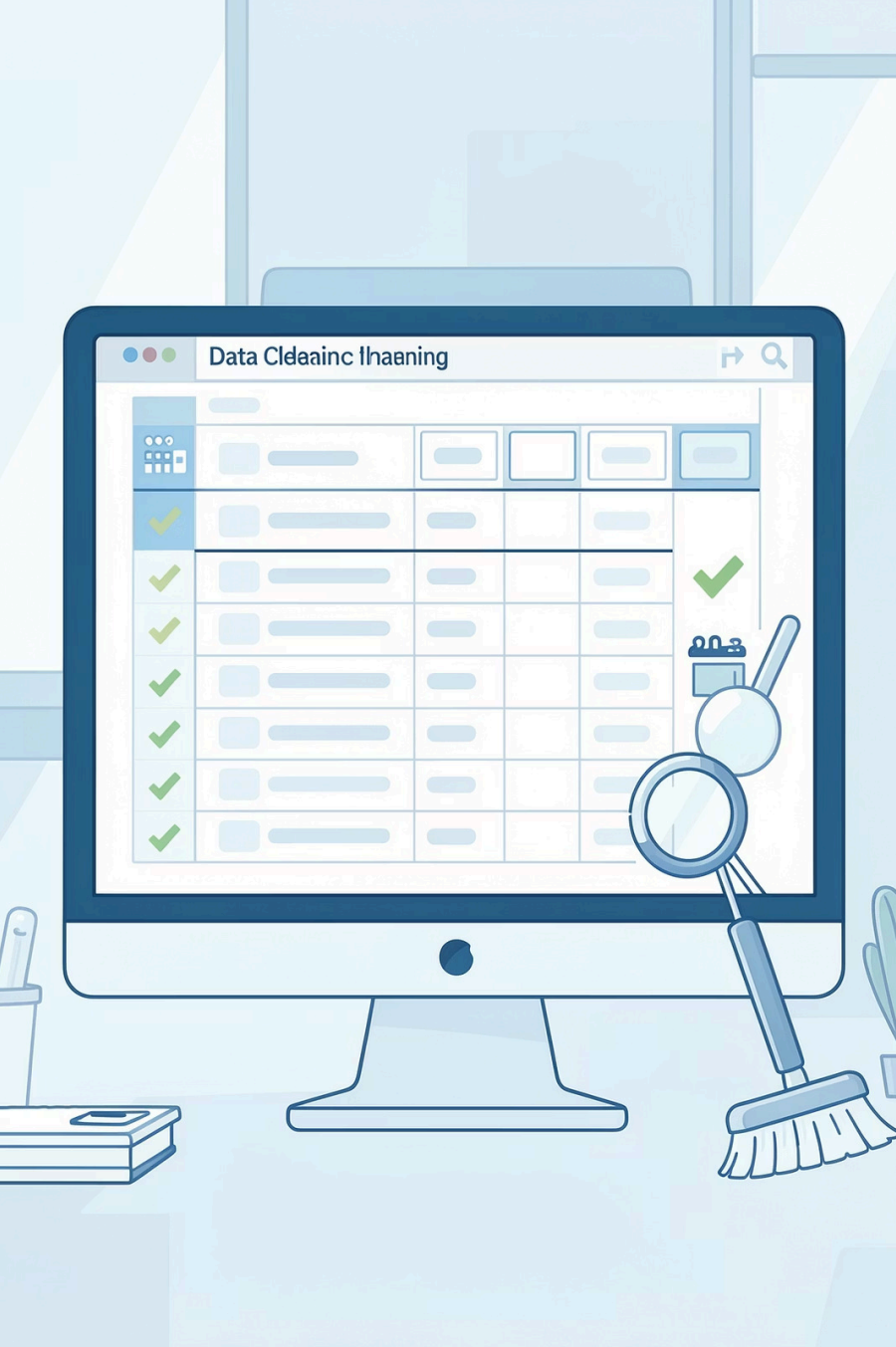
Work-Life Balance

Balance scores and overtime status



Attrition Status

Employee turnover indicator



Data Cleaning Process

Initial Data Issues

- Missing values in critical columns
- Inconsistent categorical values
- Need for feature engineering
- Data transformation required

Excel Operations

- Removed duplicate records
- Handled missing values
- Standardized categorical fields
- Structured dataset for analysis

Python Analysis (Google Colab)

Pandas was used for comprehensive data exploration and pattern identification.

01

Calculate Metrics

Total employees, attrition count, and attrition rate

02

Department Analysis

Department-wise attrition using groupby operations

03

Salary Impact

Mean income comparison of attrited employees

04

Crosstab Analysis

OverTime vs Attrition relationship study

05

Job Role Study

Job role-wise attrition pattern analysis

06

Work-Life Balance

Examination of balance impact on attrition

07

Age Groups

Age group-wise attrition analysis

08

Tenure Trends

Years at company attrition trend study

SQL Analysis

Core Metrics

Total employees, attrition count, and attrition rate calculation

Demographic Analysis

Attrition across departments, job roles, and gender

Salary Distribution

Comparison between attrited and non-attrited employees

Risk Identification

High-risk employees based on salary, overtime, and satisfaction

Dashboard Integration

SQL view creation for Power BI integration

Power BI Dashboard

Interactive Dashboard Features

- KPI cards: total employees, attrition count, attrition rate, average salary, age, and experience
- Attrition trends across education field, age groups, salary slabs, job roles, and gender
- Slicers for department, job role, and education field
- Visualizations for quick pattern identification



Key Insights



Age Factor

Higher attrition in younger age groups (26–35 years)



Salary Impact

Low salary employees show higher attrition rates



Job Roles

Lab Technician and Sales Executive have higher turnover



Overtime Effect

OverTime increases likelihood of employee attrition



Work-Life Balance

Balance and salary are major influencing factors

Real-World Applications



Reduce Attrition

Helps HR teams reduce employee attrition through targeted interventions



Workforce Planning

Supports workforce planning and retention strategies



Identify Risk Groups

Identifies high-risk employee groups for proactive measures



Improve Hiring

Improves hiring and compensation decisions



Data-Driven Management

Enables data-driven HR management

Technology Stack



Excel

Data cleaning and structuring



Python

Data exploration with Pandas



SQL

Structured insights extraction



Power BI

Interactive dashboard creation





Conclusion

This HR Analytics project demonstrates how Excel, Python, SQL, and Power BI can be used together to analyze employee data and extract meaningful insights.

Integrated Approach

Combines multiple tools for comprehensive analysis

Identifies Drivers

Helps identify key attrition drivers

Better Decisions

Supports better HR decision-making